Hints: Functional sequences, series and spaces – 1. Sequences of functions and uniform convergence

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. If $\sup|f-f_n| \leqslant \varepsilon$, then on every subinterval the oscillation of f is at most the oscillation of f_n plus 2ε, so $U(f,P) - L(f,P) \leqslant U(f_n,P) - L(f_n,P) + 2\varepsilon(b-a)$. Also f is bounded. Then $|\int f_n - \int f| \leqslant (b-a)\sup|f_n - f|$.

Hint 2. Pointwise failures: indicators of finite sets $\{r_1,\dots,r_n\}$ of rationals are integrable and converge pointwise to the Dirichlet function, which is not integrable; and $n\,\mathbf{1}_{(0,1/n)}$ (or nxe^{-nx^2}) converges to 0 with integrals not tending to 0. Arzelà's theorem says what is missing: if the f_n are uniformly bounded and the pointwise limit is Riemann integrable, then the integrals do converge.

Problem 2

Hint 1. For the example, any continuous nowhere differentiable function (Weierstrass, Takagi) is a uniform limit of C^1 functions: of its partial sums, or of polynomials by the Weierstrass approximation theorem.

Hint 2. For the positive statement, let $g = \lim f_n'$ (continuous, as a uniform limit) and pass to the limit in $f_n(x) = f_n(0) + \int_0^x f_n'(t)\,dt$; the fundamental theorem of calculus then gives $f' = g$.

Problem 3

Hint 1. If all f_n were bounded, $\varepsilon_n = 1/(n\sup|f_n|+1)$ would work; but the f_n are arbitrary, and the answer is no. The point is that $[0,1]$ is big enough to index all possible "growth rates": it has the same cardinality as the set $\mathbb{N}^{\mathbb{N}}$ of all sequences of naturals.

Hint 2. Fix a bijection $x \mapsto a^{(x)}$ from $[0,1]$ onto $\mathbb{N}^{\mathbb{N}}$ and put $f_n(x) = a_n^{(x)}$. Given any $\varepsilon_n > 0$, choose the sequence $a_n = \lceil n/\varepsilon_n \rceil$ and let x be the point coding it: then $\varepsilon_n f_n(x) \geqslant n$.

Problem 4

Hint 1. Let E be the set of positive measure where $g(x) = \lim \sin(\alpha_n x)$ exists (you may shrink E to a bounded set). First show (α_n) is bounded. If $|\alpha_{n_k}| \to \infty$, use dominated convergence together with the Riemann–Lebesgue lemma.

Hint 2. Along such a subsequence, $\int_E \sin^2(\alpha_{n_k}x)\,dx = \frac{1}{2}\int_E(1-\cos 2\alpha_{n_k}x)\,dx \to \frac{|E|}{2}$ and $\int_A \sin(\alpha_{n_k}x)\,dx \to 0$ for every measurable $A \subset E$, so $g = 0$ a.e. on E. But dominated convergence also gives $\int_E \sin^2(\alpha_{n_k}x) \to \int_E g^2 = 0$, a contradiction.

Hint 3. So (α_n) is bounded; if it had two limit points $\beta \neq \gamma$, then $\sin\beta x = \sin\gamma x$ for all $x \in E$. But $\sin\beta x - \sin\gamma x = 2\cos\frac{(\beta+\gamma)x}{2}\sin\frac{(\beta-\gamma)x}{2}$ has only countably many zeros when $\beta \neq \gamma$.

Problem 5

Hint 1. On a compact interval an equicontinuous family is uniformly equicontinuous: one δ works for all points and all n (prove this by compactness, or as in Heine–Cantor).

Hint 2. Given ε, pick finitely many points $y_1, \dots, y_k$ of the dense set forming a δ-net of $[a, b]$, and N such that $|f_n(y_i) - f_m(y_i)| < \varepsilon$ for all i and $n, m \geqslant N$. Then $|f_n(x) - f_m(x)| < 3\varepsilon$ for every x, and the Cauchy criterion finishes it.

Problem 6

Hint 1. Try $f_n(x) = \sin(2\pi n x)$. A sequence in the closed ball with a uniformly convergent subsequence would in particular have a pointwise convergent one, so it suffices to rule out pointwise convergence of any subsequence.

Hint 2. If f_{n_k} converged pointwise, then $h_k = (f_{n_k} - f_{n_{k+1}})^2$ would be continuous, bounded by 4, and tend to 0 pointwise, so $\int_0^1 h_k \to 0$ by the bounded convergence theorem (Arzelà's theorem suffices). But by orthogonality $\int_0^1 (\sin 2\pi n x - \sin 2\pi m x)^2 dx = 1$ for $n \neq m$.

Problem 7

Hint 1. Let $x \to a^+$ in $|f_n(x) - f_m(x)| \leqslant \sup|f_n - f_m|$ to get $|L_n - L_m| \leqslant \sup|f_n - f_m|$; so (L_n) is Cauchy, with limit L.

Hint 2. Then $|f(x) - L| \leqslant |f(x) - f_n(x)| + |f_n(x) - L_n| + |L_n - L|$: fix n large to make the outer terms small, then take x close to a. Counterexample for pointwise convergence: $f_n(x) = x^{1/n}$ on $(0, 1)$, with $L_n = 0$ but $f \equiv 1$.

Problem 8

Hint 1. Use tall thin spikes that move to a point where they never are: a triangle f_n supported on $[0, 2/n^3]$ with $f_n(0) = 0$ and peak value n at $1/n^3$. Then $f_n \to 0$ at every point, $\int f_n = 1/n^2$, and $\max f_n = n$.

Hint 2. If f_n were nonincreasing in n, then $\max f_n \leqslant \max f_1$; if nondecreasing with limit 0, then $f_n \leqslant 0$. Either way $\max f_n \not\to \infty$. (For a decreasing sequence Dini's theorem even gives uniform convergence.)

Problem 9

Hint 1. Polynomials of degree at most d are determined by their values at $d + 1$ fixed nodes $t_0, \dots, t_d$, and the Lagrange formula expresses them linearly through these values.

Hint 2. If $p_n \to f$ uniformly, the values $p_n(t_i)$ converge to $f(t_i)$, so by the Lagrange formula p_n converges uniformly (on the bounded interval) to the interpolation polynomial p of the values $f(t_i)$, hence $f = p$. For the second part apply the Weierstrass approximation theorem to a continuous nowhere differentiable function.

Problem 10

Hint 1. The answer is no. Baire's theorem only gives, for each ε, an interval on which $\sup_{n \geqslant N} |f_n - f| \leqslant \varepsilon$, and these intervals may shrink as $\varepsilon \to 0$; a counterexample has to exploit exactly this.

Hint 2. Let $\varphi_n(t) = \max(0, 1 - |nt - 1|)$, a spike of height 1 on $[0, 2/n]$; $\varphi_n \to 0$ pointwise. Enumerate the rationals r_k of $[0, 1]$ and set $f_n(x) = \sum_k 2^{-k} \varphi_n(x - r_k)$.

Hint 3. Check: f_n is continuous with values in $[0, 1]$ (M-test), $f_n(x) \to 0$ for each x (each term tends to 0 and is dominated by 2^{-k}), but every interval contains some r_k in its interior, and then $\sup f_n \geqslant 2^{-k}$ on that interval for all large n.